HAPPYROBOT CHALLENGE · TUM.AI MAKEATHON 2026

Cold outbound is dead in Germany.

So we inverted the direction.

Three AI agents make the prospect call us.

lease·a·kitchen — Germany's first appliance-as-a-service operator for senior-living, built on Bosch All-in+.

Team **undeterministic tornado** · TUM.ai Makeathon 2026

THE PROBLEM

German law killed B2B outbound.

§7 UWG

Cold calls need consent. AI voice calls need *explicit* consent — always.

<1%

Cold email response rate. LinkedIn and WhatsApp follow the same curve — and violate ToS.

300k €

Max. fine per violation. One careless campaign ends the company.

Meanwhile, buyers publish intent constantly — **openings, funding, job posts, permits** — that no one converts into timely, legal outreach.

THE IDEA

Stop chasing. Listen — then invite.

Every opening, funding round, or tender leaves a public trace. We read those traces, act on the strongest one, and let the prospect open the door.

OLD WAY

Push a cold message to thousands. Pray for 1 %. Lawyer's number on speed-dial.

→ INVERT

OUR WAY

Read public signals. Send one postcard to the right person. They scan, they call us.

A REAL EXAMPLE — WHAT JACK SAW THIS WEEK

PUBLIC SIGNAL

SEEN BY JACK

sueddeutsche.de · Press · April 2026

score 84

Augustinum Gruppe launches 50-year anniversary programme — 180 apartments refurbished across Munich by 2027

"...the refurbishment covers kitchens, bathrooms and appliances across *Stiftsbogen*, *Nord* and *Neufriedenheim* residences. Programme lead: **Sabine Fischer**, Hausleitung..."

`motivation_string` = **simplify** · ICP gate ✓ **passed** · triggers postcard to `L-10`

OUR SOLUTION

Three agents. Each one ruthlessly scoped.

Postal is §7-free. The prospect opens the door. Then three specialists do exactly one job each.

JACK · SCOUT

Hunts public buying signals.

"Who's about to need this — and why?"

Sources — KfW funding, building permits, public tenders, expansion press

Scores — 6 weighted dimensions, 0–100, with hard ICP gate (no commercial kitchens)

Picks the hook — Simplify · Scale · Optimize · Circular, by decision-maker role

Triggers — personalised postcard with QR to a hook-tuned landing page

KATE · CALLER

Dials only when invited.

"Tell me about your project — I'll suggest the fit."

Triggered — by the QR-scan + consent on the landing page (UWG §7 satisfied)

Discloses — AI in the first sentence (EU AI Act Art. 50)

Discovers + configures — Leader / Profi / Top Feature mix in 4 minutes

Hands off — qualified envelope written into Twin → Otto fires

OTTO · CLOSER

Builds the proposal. Sends it.

"Reads the call, prices the bundle, hits send."

Reads — full Twin row (bundle mix, term, contact channel)

Computes — €/month × term, with the BSH Leader/Profi/Top Feature rates

Drafts + sends — tailored email body via the Outlook integration

Updates — stage → "offered", offer_sent_at → now

`Signal → Postcard → QR + Consent → Voice qualification → Offer email`

What Ms. Fischer actually sees.

Three touchpoints. Real layouts from the demo environment — nothing mocked.

① THE POSTCARD

Dear Ms. Fischer,

your **50th-anniversary refurbishment programme** across the Munich residences — 180 apartments through 2027 — is the perfect moment to rethink the appliance setup.

We deliver, install, maintain and replace. **48-hour express swap**, one invoice, one contact.

2-minute check.
Scan the QR → tailored proposal.

Ms.
Sabine Fischer
Augustinum Gruppe
Stiftsbogen 74
81375 München

Best regards,
Frank Eberle
Sales Director · lease·a·kitchen

DIN A6, personalised by Jack. QR encodes `?id=L-10`.

→ SCAN

② THE LANDING PAGE

lease·a·kitchen lease-a-kitchen.de

SIMPLIFY · ONE PARTNER, ONE INVOICE

For **there** at your organisation

Your facility team never chases a repair invoice again.

One partner, one contract, one monthly rate. Delivery, installation, repair, replacement, end-of-life — all bundled. Your team gets on with the resident-facing work that actually matters.

48 h	1	0
EXPRESS REPLACEMENT SLA	CONTRACT · CONTACT · INVOICE	WARRANTY PAPERWORK

How it works. Staff or resident reports an issue. BSH dispatches — repair or swap, within 48 h at the door or 72 h on site. Maintenance, parts, consumables: all in the monthly rate.

WHY WE'RE REACHING OUT

We saw **your recent public announcement**. Based on that, we think simpler service for your team is the conversation worth having. Kate can explain in 15 minutes and pull together concrete numbers for 120 units.

Talk to Kate · 15 min

§7 UWG	AI Act 50	GDPR
compliant outreach	disclosed on call	fully logged

operated by orgaloom GmbH · Munich
Privacy · Imprint · Terms

Hook = `simplify`. One tap books Kate.

→ OPT-IN

③ OTTO'S OFFER EMAIL

Outlook · Sent

To s.fischer@augustinum.de
From frank.eberle@lease-a-kitchen.de

Your 180-apartment refurbishment — tailored proposal

Dear Ms. Fischer,

following our call, here is the configuration we discussed for the 50th-anniversary programme across the Munich residences:

180 × Profi (Serie 4–6)	58 €
term	60 months
Monthly, all-in	**10,440 €**
incl. delivery · maintenance · 48 h swap · end-of-life	

Valid for 14 days. One click below locks the terms; no obligation.

Accept proposal →

Best regards,
Frank Eberle · Sales Director

Generated by Otto in ~60 s after Kate's call ends. Sent via Outlook.

Total elapsed: postcard mailed → signed offer in the inbox ≈ 4 working days.

THE COST GATE

Jack protects the budget — and the homologation.

A postcard costs ~1,20 €. Jack sends one only if ICP fits (household-pattern deployment, never commercial kitchens) *and* the deal value justifies the spend.

SIX SCORING DIMENSIONS

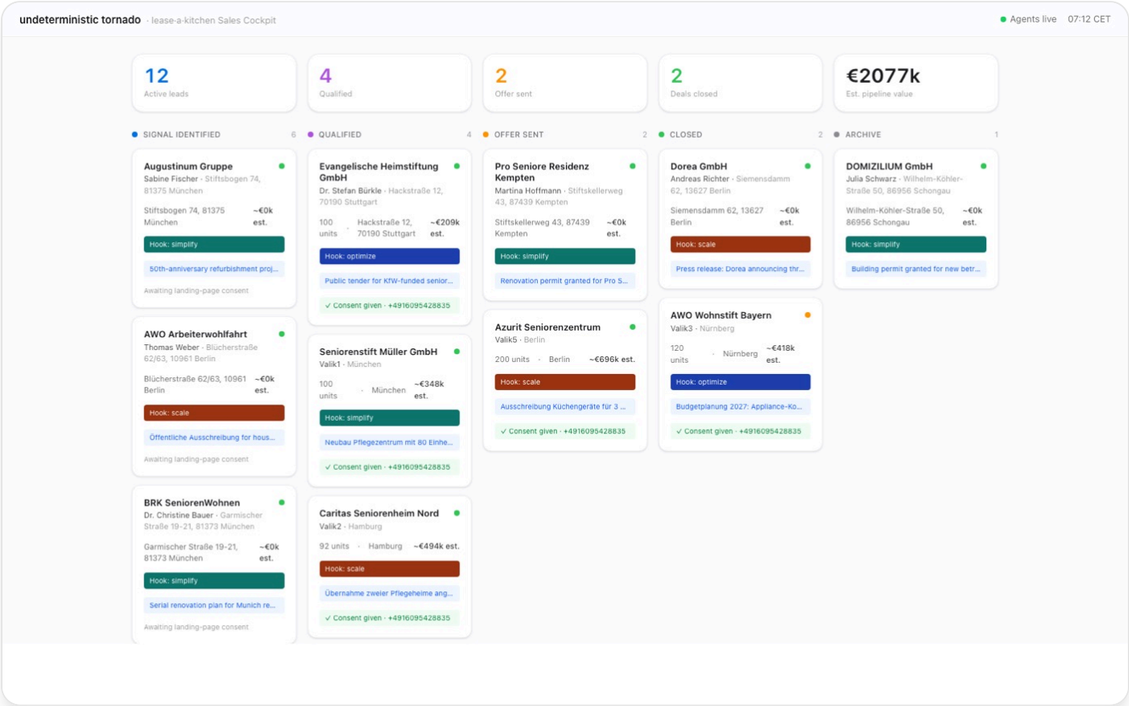
Signal Strength	30%
Deal Size	25%
Inventory Fit	20%
Contact Data Quality	15%
Timing / Urgency	10%
Risk / Exclusion	KILL

```
# jack · top 3 · today

diakonie münchen                                  87.3 PREMIUM
KfW funded · 120 units · per-apt deploy · DM named

stuvent dorm gmbh                                 74.1 STANDARD
Berlin · 80-unit student housing · permit granted

stadthotel group                                  68.8 STANDARD
Aparthotel rollout · in-room fridges · role only

suppressed: 14 (homologation · insolvency · private)
watchlist: 28 (re-score in 30d)
```

LIVE DEMO · COCKPIT AT `MAKEATHONTUMAI.VERCEL.APP`

Let's call Kate.

One real phone call. Watch the cockpit update as she talks.



Real Twin data · live pipeline · polls every 4 s

WHAT YOU'LL SEE

1. **A call with Kate.** Live, on stage. AI disclosure in sentence one, then discovery + bundle configuration in plain language.
2. **The cockpit follows along.** Lead card moves through the pipeline as Kate fills in facility, units, term, bundle mix.
3. **Otto takes over.** Reads the row, prices the bundles, sends the offer email — while we're still on the slide.

We've prepared three personas (L-1, L-3, L-10) you can pick from. Everything you see is running against the real Twin database.

STACK · Built on HappyRobot

No glue code we didn't have to write.

WHAT HAPPYROBOT GIVES US

Twin · managed Postgres
One `leads` table. All three agents read + write via Read / `Write` nodes. Cockpit reads via REST.

Native Contacts + Memories
Auto-dedup on phone/email. Per-call interaction history. No CRM glue.

Webhook trigger + POST
Bearer-auth endpoints kick agents off. Outbound POSTs back to the cockpit close the loop.

Custom Code (Python)
Otto's bundle math: `leader×42 + profi×58 + top×80`. 18 lines.

Outbound Voice Agent
Kate's call. STT → LLM → TTS pipeline, @variable interpolation, AI Act Art. 50 disclosure.

AI Extract · strict JSON Schema
Kate's transcript → 16 typed fields with `additionalProperties: false`. No regex, no parsing.

Outlook integration
Otto sends the proposal via the org's authorised mailbox — no SendGrid setup.

Runs · transcripts · audit
Full execution timeline per call. Latency breakdown for STT/LLM/TTS.

WHAT WE ADDED ON TOP

Cockpit (Next.js on Vercel)
Kanban, KPI bar, lead detail with transcript, polls Twin every 4 s.

Hook-tuned landing page
One `/api/l?id=…` endpoint, four motivation variants.

DIN A6 postcard pipeline
QR per prospect, Deutsche-Post-compliant layout.

Three agent prompts + scoring
Jack scoring rubric, Kate's consultative configurator, Otto's offer template.

~1.2k lines of TS · **~700 lines** of agent prompts · **7 HR workflow nodes**.

MOAT

§7 UWG-compliant by design. Real Pipedrive lead source. BSH commercial-agent contract operating today. Three roles, four channels, one row per lead — the system competitors can't shortcut.

THANK YOU

This isn't a hackathon demo.
It's my sales team.

I run **lease·a·kitchen**. The §7-UWG problem is mine every morning. I'm building this. Watch us ship.

lease-a-kitchen.de Follow along →

Team undeterministic tornado · Built on HappyRobot · github.com/ValikProgrammer/Makeathon-TUM-AI